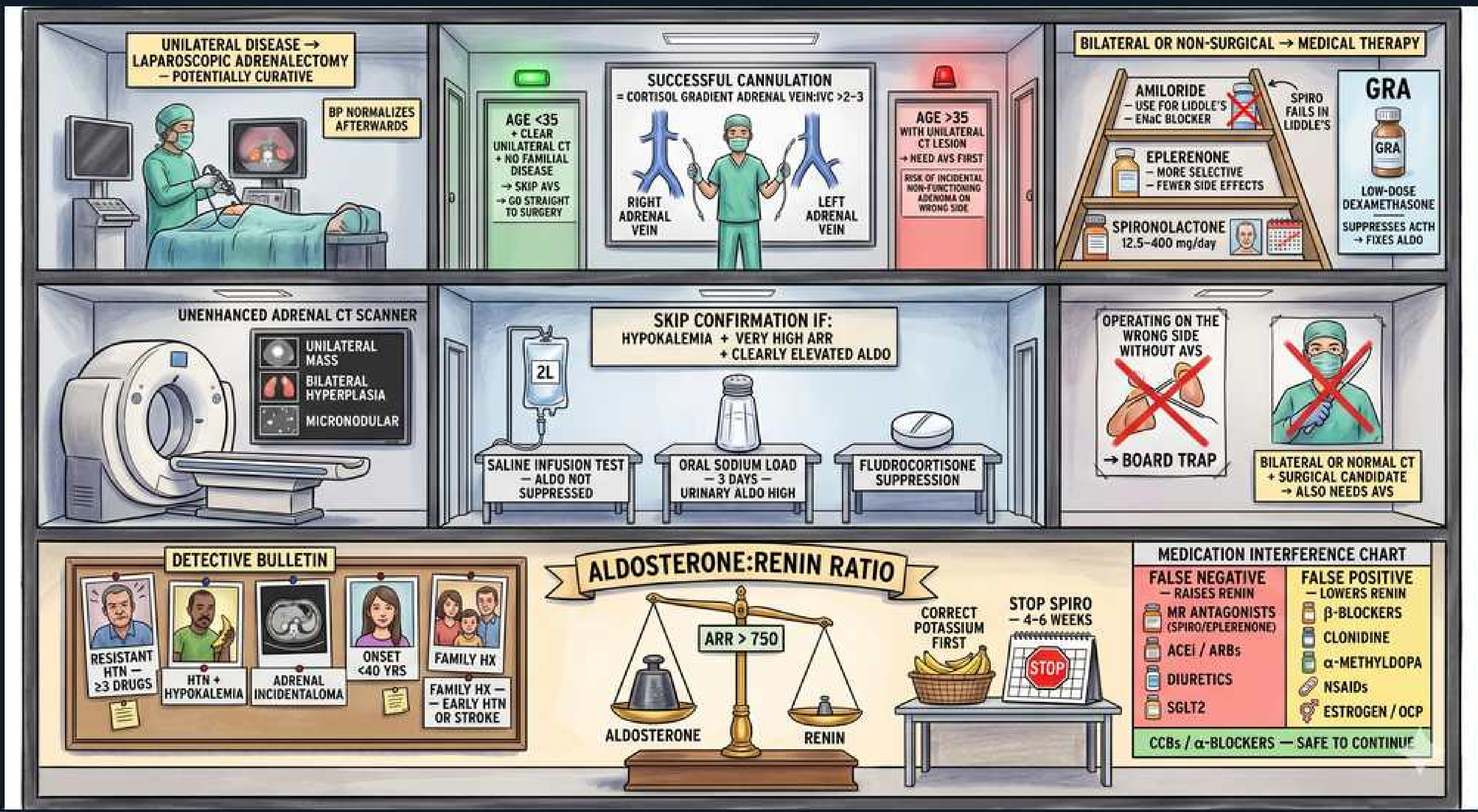
Primary Aldosteronism — The ARR Detective Lab



1. Who to screen

WHAT YOU SEE

DETECTIVE BULLETIN: resistant HTN ≥3 drugs, HTN + hypokalemia, adrenal incidentaloma, onset <40, family hx

WHAT IT ENCODES

Screen for primary aldosteronism in: resistant HTN (uncontrolled on ≥3 drugs including a diuretic, or controlled needing ≥4), HTN with spontaneous or diuretic-induced hypokalemia, HTN with an adrenal incidentaloma, young-onset HTN (<40), HTN with OSA, and HTN with a family history of early HTN or stroke <40. Up to ~20% of resistant HTN is primary aldosteronism.

BOARD TRAP

Only screening hypokalemic patients — MOST patients with primary aldo are NORMOKALEMIC, so normal potassium does not exclude it.

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2. ARR screening test

WHAT YOU SEE

Central ALDOSTERONE:RENIN RATIO scale, ARR > 750

WHAT IT ENCODES

The screening test is the plasma ALDOSTERONE-to-RENIN RATIO (ARR): a HIGH aldosterone combined with a SUPPRESSED renin gives a high ratio (the scene shows ARR > 750 with PRA in ng/mL/h; cutoffs depend on units). The combination matters — you need both an elevated aldosterone (commonly >10-15 ng/dL) AND low renin, not the ratio alone. Draw mid-morning, seated, after the patient has been ambulatory.

BOARD TRAP

Acting on a high ratio driven purely by very LOW renin when aldosterone is not actually elevated — a high ARR with a low absolute aldosterone is usually not primary aldo. Always check the aldosterone level, not just the ratio.

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3. Correct potassium first

WHAT YOU SEE

Note CORRECT POTASSIUM FIRST near the scale

WHAT IT ENCODES

Correct hypokalemia BEFORE measuring the ARR. Low potassium suppresses aldosterone secretion (K is a direct stimulus to the zona glomerulosa), which can lower the aldosterone and produce a FALSE-NEGATIVE screen. Also ensure the patient is not sodium-restricted.

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4. Stop interfering drugs

WHAT YOU SEE

STOP SPIRO 4-6 weeks, red STOP sign

WHAT IT ENCODES

Stop mineralocorticoid-receptor antagonists — spironolactone and eplerenone (and high-dose amiloride) — for 4-6 weeks before testing, because they raise renin and invalidate the ARR. If BP allows, also withhold ACEi/ARB and diuretics. Bridge BP with agents that minimally affect the ARR (see chart).

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5. Medication interference chart

WHAT YOU SEE

Right chart: false negatives vs false positives; CCBs/alpha-blockers safe

WHAT IT ENCODES

FALSE NEGATIVES (raise renin, lower the ratio): MR antagonists, diuretics, ACE inhibitors, ARBs, dihydropyridine CCBs, SGLT2 inhibitors. FALSE POSITIVES (lower renin, raise the ratio): beta-blockers, central agonists (clonidine, alpha-methyldopa), NSAIDs, and estrogen/OCP. Drugs SAFE to continue for BP control: non-dihydropyridine CCBs (verapamil), alpha-blockers (doxazosin), and hydralazine.

BOARD TRAP

Drawing the ARR on a beta-blocker (suppresses renin → false POSITIVE) or on an ACEi/ARB or MRA (raises renin → false NEGATIVE). Switch to verapamil/alpha-blocker first.

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6. Skip confirmation if classic

WHAT YOU SEE

SKIP CONFIRMATION IF: hypokalemia + very high ARR + clearly elevated aldo

WHAT IT ENCODES

Confirmatory testing can be SKIPPED when the picture is unambiguous: spontaneous hypokalemia + a clearly elevated aldosterone + an undetectable/very suppressed renin + a very high ARR. In that case proceed directly to subtype evaluation (CT ± AVS).

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7. Confirmatory tests

WHAT YOU SEE

Saline infusion, oral sodium load, fludrocortisone suppression — aldo NOT suppressed

WHAT IT ENCODES

Otherwise CONFIRM autonomy with a suppression test that demonstrates aldosterone fails to suppress with sodium/volume loading: IV saline infusion test (2 L over 4 h — aldosterone stays high), oral sodium loading (3 days, high urinary aldosterone), or fludrocortisone suppression. In a normal person volume expansion suppresses aldosterone; in primary aldo it does not.

BOARD TRAP

Expecting aldosterone to fall — failure to suppress is the POSITIVE/confirmatory result. (Captopril challenge is an alternative if saline is risky, e.g., heart failure.)

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8. Unenhanced CT

WHAT YOU SEE

CT scanner: unilateral mass, bilateral hyperplasia, micronodular

WHAT IT ENCODES

After biochemical confirmation, do an adrenal CT (thin-cut, non-contrast) to look for a unilateral adenoma, bilateral hyperplasia, micronodular change, or to exclude a large mass suspicious for carcinoma. CT defines anatomy but does NOT reliably establish which side is the functional source.

BOARD TRAP

Trusting CT alone to lateralize — non-functioning incidentalomas are common, and CT misses small functional micro-adenomas or mislabels bilateral hyperplasia, leading to wrong-side surgery.

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9. AVS lateralizes

WHAT YOU SEE

OPERATING ON THE WRONG SIDE WITHOUT AVS — BOARD TRAP, red X

WHAT IT ENCODES

Adrenal vein sampling (AVS) is the GOLD STANDARD to lateralize and is required in most patients ≥35 before adrenalectomy, because CT and the true functional source disagree in a substantial fraction of cases. AVS distinguishes a surgically curable unilateral source from bilateral hyperplasia that needs medical therapy.

BOARD TRAP

Operating based on CT findings alone without AVS — the single most tested error in this workup. CT-AVS discordance is common.

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10. AVS cannulation

WHAT YOU SEE

SUCCESSFUL CANNULATION, cortisol gradient adrenal vein/IVC >2-3

WHAT IT ENCODES

AVS interpretation has two steps. (1) Selectivity/confirm cannulation: the adrenal-vein-to-IVC CORTISOL ratio must exceed ~2 (unstimulated) or ~3-5 (with cosyntropin) on each side to prove the catheter is actually in the adrenal vein. (2) Lateralization: compare cortisol-corrected aldosterone between sides; a dominant side identifies the unilateral source.

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11. Young + unilateral → skip AVS

WHAT YOU SEE

AGE <35, unilateral CT lesion, no familial, skip AVS go to surgery

WHAT IT ENCODES

AVS may be omitted in patients <35 years with a clear solitary unilateral adenoma on CT, a normal contralateral gland, and no family history — because at that age non-functioning incidentalomas are rare, so CT is reliable. These patients can go straight to adrenalectomy.

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12. Unilateral → adrenalectomy

WHAT YOU SEE

UNILATERAL DISEASE laparoscopic adrenalectomy, potentially curative, BP normalizes

WHAT IT ENCODES

Confirmed UNILATERAL disease (aldosterone-producing adenoma or unilateral hyperplasia) → laparoscopic adrenalectomy. This is potentially CURATIVE: hypokalemia resolves and BP improves or normalizes in the majority, with the best response in younger patients with shorter HTN duration.

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13. Bilateral → MRA

WHAT YOU SEE

BILATERAL/NON-SURGICAL MEDICAL THERAPY: spironolactone, eplerenone, amiloride, GRA dexamethasone

WHAT IT ENCODES

BILATERAL hyperplasia or any non-surgical candidate → lifelong MR antagonist: spironolactone first-line (12.5-400 mg, but watch gynecomastia/menstrual irregularity) or eplerenone if anti-androgen effects are a problem. Amiloride (ENaC blocker) is an alternative. For glucocorticoid-remediable aldosteronism (GRA / familial hyperaldosteronism type 1, CYP11B1-CYP11B2 chimeric gene), use low-dose DEXAMETHASONE to suppress ACTH-driven aldosterone.

BOARD TRAP

Forgetting GRA in a young patient with a family history of early HTN/hemorrhagic stroke — it is treated with low-dose glucocorticoid, not just an MRA.

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14. Adrenalectomy operating room

WHAT YOU SEE

Top-left panel: surgeons performing laparoscopic adrenalectomy, BP normalizes afterward

WHAT IT ENCODES

The laparoscopic adrenalectomy panel illustrates the curative arm for unilateral disease. Post-op, continue monitoring K and BP; the suppressed contralateral zona glomerulosa can cause transient HYPERKALEMIA, so stop K supplements/MRAs and watch potassium after surgery.

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15. Saline/sodium load confirmatory props

WHAT YOU SEE

2L saline bag, oral sodium load and fludrocortisone bottles in the center

WHAT IT ENCODES

The 2-L saline bag and salt/fludrocortisone props are the physical embodiment of volume loading: each delivers a sodium/volume challenge that should switch OFF aldosterone in a normal person. Persistently high aldosterone despite the load confirms autonomous secretion — the definition of primary aldosteronism.
